FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2019

PAPER –1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, C

Sol. Potential of sphere

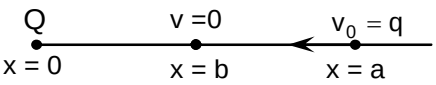
$$= \frac{kQ}{R} + \frac{kq}{a} = 0 \Rightarrow Q = \frac{q}{a}R$$

Current flowing to the sphere $I = \frac{dQ}{dt} = \frac{-q}{a}\frac{dR}{dt} = -\frac{qv}{a}$

So current flowing away from sphere $= -I = \frac{qv}{a}$

2. A, C

Sol.



Applying conservation of energy:

$$\frac{1}{2}mv_0^2 + \frac{kQq}{a} = \frac{kQq}{b}$$

solve to get $\frac{b}{a} = \frac{6}{7}$

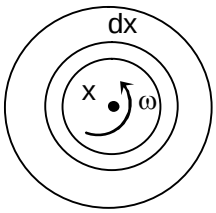
Again apply conservation of energy between x = a and $x = \infty$.

$$\frac{1}{2}mv_0^2 + \frac{kQq}{a} = \frac{1}{2}mv^2$$

Solve to get $v = \sqrt{7}v_0$

3. A, D

Sol. Consider a ring of radius x and thickness dx.



Equivalent current in this ring $= \frac{\omega}{2\pi} \times$ charge on ring

$$= \frac{\omega}{2\pi} \times (2\pi \times dx) \frac{Q}{\pi R^2}$$

dB (due to this ring) $= \frac{\mu_0}{2x}\left(\frac{\omega}{2\pi}\frac{2xQ}{R^2}dx\right)$

$$\therefore B = \int_0^R \frac{\mu_0 \omega}{2\pi}\frac{Q}{R^2}dx. = \frac{\mu_0 \omega Q}{2\pi R^2}.R = \frac{\mu_0 \omega Q}{2\pi R}$$

4. C, D

Sol. For cylinder;

$$B = \frac{\mu_0 ir}{2\pi R^2}; r < a$$

$$= \frac{\mu_0 i}{2\pi r}; r \geq a$$

We can consider the given cylinder as a combination to two cylinders. One of radius R carrying current I in one direction and other of radius R/2 carrying I/3 in both directions.

At point A:

$$B = \frac{\mu_0 (I/3)}{2\pi (R/2)} + 0 \frac{\mu_0 I}{3\pi R}$$

At point B:

$$B = \frac{\mu_0}{2}\left(\frac{4I/3}{\pi R^2}\right)\left(\frac{R}{2}\right) + 0 = \frac{\mu_0 I}{3\pi R}$$

5. B, C, D

Sol. (A) The frequency of oscillation of the circuit is $f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{\sqrt{(0.100\ H)\left(1.00\times10^{-6}F\right)}} = 503\ Hz$

(B) The maximum charge on the capacitor is $Q = C\varepsilon = \left(1.00\times10^{-6}F\right)(12.0\ V) = 12.0\ \mu C$

To find the maximum current I, we equate $\frac{1}{2}C\varepsilon^2 = \frac{1}{2}LI^2$,

Then solve for I, to obtain

$$I_i = \varepsilon\sqrt{\frac{C}{L}} = 12.0\ C\sqrt{\frac{1.00\times10^{-6}F}{0.100H}} = 37.9\ mA$$

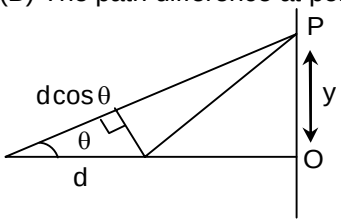
(D) The total energy the circuit possesses at t = 3.00 s and at all

$$U = \frac{1}{2}C\varepsilon^2 = \frac{1}{2}\left(1.00\times10^{-6}F\right)(12.0\ V)^2 = 72.0\mu J$$

6. A, B, C, D

Sol. (A) The line joining point source S_1 and S_2 is perpendicular to screen. Hence, shape of fringe is circular.

(B) The path difference at point P is



$\Delta x = d\cos\theta$

For the minima. $\Delta x = \frac{D}{2}$

$\therefore d\cos\theta = \frac{D}{2}$ or $\cos\theta = \frac{D}{2d}$

or $\frac{y}{\sqrt{D^2+y^2}} = \frac{D}{2d}$ or $\frac{y^2}{D^2+y^2} = \frac{D^2}{4d^2} = \frac{1}{4\times 4.8^2}$

or $92.16y^3 = 1 + y^2$ or $91.16y^2 = 1$

$\therefore y = 0.1\,m = 10\,cm$

(C) $\Delta x = d\cos\theta = (2m-1)\frac{D}{2}$ $\therefore m = -4,-3,-2,-1,0,1,2,3,4,5$

Thus total number of minima is 10.

(D) The path difference at point O is $\Delta x = d = 4.8D$. Hence, it is neither a point of maxima nor a point of minima.

7. A, D

Sol. In the time light takes to go from the wheel to the mirror and back again, the wheel turns through an angle of

$$\theta = \frac{2\pi}{500} = \frac{\pi}{250}\,\text{rad}$$

That time is $t = \frac{2l}{C} = \frac{2(500m)}{3\times 10^8\ m/s} = \frac{10}{3}\times 10^{-6}\,s = \frac{10}{3}\mu s$

So the angular velocity of the wheel is

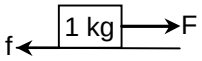
$$\omega = \frac{\theta}{t} = \frac{\frac{\pi}{250}\,\text{rad}}{(10/3)\times 10^{-6}\,s} = 1200\pi\ \text{rad/s}$$

If r is the radius of the wheel, the linear speed of a point on its rim is

$v = \omega r = (1200\pi\ \text{rad/s})\,(0.050\ \text{m}) = 60\pi\ \text{m/s}$.

8. A

Sol.



$F = f_{max}$ at equilibrium

$= \mu N = \mu mg$

9. B

Sol. $F - f = MA$

$A = \frac{F-f}{M}$

10. C

11. D

Sol. $\tau = I\alpha \Rightarrow \alpha = \frac{\tau}{I}$

I = moment of inertia

12. B

Sol. $K_{system} = K_1 + K_2 + K_3$

When $V_{max} \Rightarrow KE_{max}$

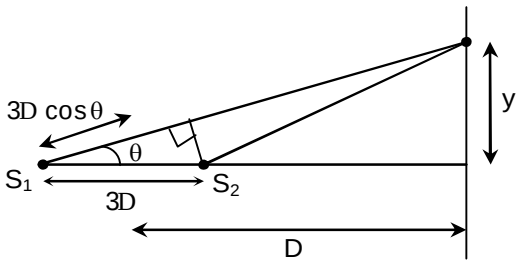
13. A

Sol. Angular momentum $= I\omega$

SECTION – C

14. 7

Sol.



Path difference = $3D\cos\theta = 2D \Rightarrow \cos\theta = 2/3$

$y = D\tan\theta = \frac{D\sqrt{5}}{2} \Rightarrow m + n = 5 + 2 = 7$

15. 5

Sol. Here, $\Delta x = d\sin\theta - (SS_2 - SS_1)$

$= d\sin\theta - 40\ \mu m = m\lambda$

or $d\sin\theta = m\lambda + 40\ \mu m$

$\because -1 < \sin\theta < 1$

or $-1 < \frac{m \times 100 \times 10^{-9}}{10 \times 10^{-6}} + 40 \times 10^{-6} < 1$

or $-2500 < 1000m + 1 < 2500$

or $-2.5 < m < 2.5$

or $\therefore m = -2, -1, 0, 1, 2$

16. 2

Sol. Let ground state energy (in eV) be E_1

Then from the given condition

$E_{2n} - E_1 = 204$ eV or $\frac{E_1}{4n^2} - E_1 = 204$ eV

$\Rightarrow E_1\left(\frac{1}{4n^2} - 1\right) = 204$ eV .....(i)

and $E_{2n} - E_n = 40.8$ eV

$\Rightarrow \frac{E_1}{4n^2} - \frac{E_1}{n^2} = E_1\left(-\frac{3}{4n^2}\right) = 40.8$ eV ....(ii)

From euqation (i) and (ii), $\frac{1-\frac{1}{4n^2}}{\frac{3}{4n^2}} = 5 \Rightarrow n = 2$

17. 8

Sol. Acceleration $a \propto \frac{v^2}{r}$

where $v \propto \frac{Z}{n}$ and $r \propto \frac{n^2}{Z} \Rightarrow a \propto \frac{Z^3}{n^4}$

Since both are in ground state i.e., n = 1

so $a \propto Z^3 \Rightarrow \frac{a_{He^+}}{a_H} = \left(\frac{Z_{He^+}}{Z_H}\right)^3 = \left(\frac{2}{1}\right)^3 = \frac{8}{1}$

18. 2

Sol. $\frac{dN}{dt} = t^2 - \lambda N$

For $\frac{dN}{dt}$ to be minimum; $\frac{d^2N}{dt^2} = 0$

$\Rightarrow \frac{d^2N}{dt^2} = 2t - \lambda\frac{dN}{dt} = 2t - \lambda\left(t^2 - \lambda N\right) = 0$

or $N = \frac{2t_0 - \lambda t_0^2}{\lambda^2}$

Chemistry

PART – II

SECTION – A

19. A, B, C, D

Sol. a is $CuFeS_2$, b is Cu_2S, c is Cu_2O, d is Cu_3FeS_3

20. A, B, C, D

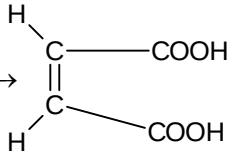
Sol. (a) In BCl_3. 2P orbital of B will overlap with lone pair of Lewis base while in $AlCl_3$ it is 3 P orbital of Al.
(b) back bonding is reason
(c) bond strength decreases so reducing power increases
(d) bond angle in PH_2 : 93.5^o
Bond angle in NH_3 : 107°
Bond angle in H_2O : 104.5^o
Bond angle in $(C_2H_5)_2O$ is greater than $109^o.28$

21. C, D

Sol. $B_2O_3 + H_2O \longrightarrow H_3BO_3$

$N_2O_5 + H_2O \longrightarrow HNO_3$

$P_4O_{10} + H_2O \longrightarrow H_3PO_4$

Maleic anhydride + $H_2O \longrightarrow$ 

Maleic acids

H_3PO_4 & maleic acid are polyprotic acids.

22. A, B, C, D

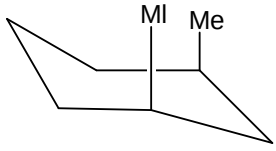
Sol. All are exothermic reactions.

23. A, B, C

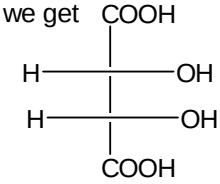
Sol. Xanthoproteic test is given by such amino acids which contains aromatic groups that derivatives of benzene.

24. A, B, C

25. B, D

Sol. (B) 

2 chiral carbon but has POS

(D) Do even no of exchanges at top chiral carbon we get has POS with 2 chiral carbons. 

26. A
27. C
28. A

29. D

Sol. $P \rightarrow$ no reaction $y \Rightarrow H_3PO_3$
$P + Q \Rightarrow HIO_2$ and HIO_3
$R \Rightarrow HClO_4$
Reaction q is vigorous

30. A
31. A

SECTION – C

32. 3

Sol. (Z+A) =
X = Cis – But – 2 ene
Y = Trans But – 2 ene
Z = Racemic mixture
A = Meso compound

33. 3

Sol. Let the rate law es Rate = $k\left[P_{NO_2}\right]^x\left[P_{H_2}\right]^y$ By putting the different values and soling And soling we will get

$$Rate = k\left[P_{NO_2}\right]^2\left[P_{H_2}\right]^1$$

Order = 2 + 1 = 3

34. 2

Sol. $In\, k = 78.09 - \dfrac{42075}{500} = -6.06$

$\log k = 2.344 \times 10^{-3}\ \min^{-1}$

$t_{1/4} = \dfrac{2.303}{2.344 \times 10^{-3}}\, by 1.333$

$= 123.06\ \min = 2\ hrc$

35. 2

Sol. Only primary amine will give carbylamines test.

36. 3

Sol. $-OCH_3$ is not ambident nucleophlice

Mathematics

PART – III

SECTION – A

37. B, C, D

Sol. $\because \lim\limits_{n\to\infty}\sum\limits_{K=0}^{k=n}\dfrac{{}^{n}C_{K}}{n^{K}(K+3)}=\int\limits_{0}^{1}x^{2}\,e^{x}\,d_{k}$

$=x^{2}e^{x}\,|_{0}^{1}-\int\limits_{0}^{1}2x.e^{x}\,dx$

$=e-2\left[xe^{x}\right]_{0}^{1}-\int\limits_{0}^{1}e^{x}dx$

$=e-2\left(e-\left(e-1\right)\right)=e-2 \quad \therefore \begin{cases} a= & -2 \\ b= & 1 \end{cases}$

38. A, C

Sol. $\because f'(x)=2xe^{-x}$

$\Rightarrow f(x)=-2e^{-x}(x+1)+c$

$f(0)=-2+c\Rightarrow c=2$

$f(x)=2-2.e^{-x}(x+1)$

$e^{x}f(x)=2e^{x}-2x-2$

$\Rightarrow e^{x}\left(f(x)-2\right)=-2(x+1)$

$\Rightarrow \left(2-f(x)\right)e^{x}=2(1+x)$

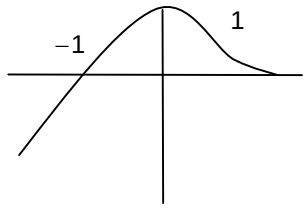
$x=\ln 5$

$\Rightarrow \left(2-f(\ln 5)\right)5=2x(1+\ln 5)$

$g(x)=(1+x)e^{-x}$

$g'(x)(1+x)\left(-e^{-x}\right)+1\times\left(e^{-x}\right)$

$=e^{-x}\left(1-\times-x\right)=-xe^{-x}$



39. A, C

Sol. $\because \alpha=\sqrt{2}-1=\alpha^{2}=3-2\sqrt{2}$

& $\beta=2\sqrt{2}-2 \quad \therefore \alpha^{2}+\beta=1$

$\beta=2\alpha$

40. A, B, C, D

Sol. $x=\dfrac{12}{15}\times\dfrac{1}{\left(1+\dfrac{1}{x^{2}+y^{2}}\right)}$ ..... (i)

and $y=\dfrac{4}{5}\times\left(\dfrac{1}{1-\dfrac{1}{x^{2}+y^{2}}}\right)$ ..... (ii)

Put $x^2+y^2=\frac{1}{u}$ and add $(i)^2$ and $(ii)^2$

$\Rightarrow \frac{1}{u}=\left(\frac{4}{5}\right)^2\left(\frac{9}{(1+4)^2}+\frac{1}{(1-4)^2}\right)$

$\Rightarrow 25\left(u^2-2+\frac{1}{u^2}\right)=32\left\{5\left(4+\frac{1}{4}\right)-8\right\}$

Put $u+\frac{1}{u}=v \Rightarrow v=\frac{6}{5}$ or $\frac{26}{5}$

$\Rightarrow 4=5$ or $\frac{1}{5}$

$\Rightarrow (x, y)=(2, 1)$ or $\left(\frac{2}{5}, \frac{-1}{5}\right)$

41. A, C

Sol. $\because \quad x\to 0^- \Rightarrow \{x\}=1+x$

$\therefore \quad \lim\limits_{x\to 0^-} f(x)=\lim\limits_{x\to 0}\frac{\sin^{-1}(-x)\cos^{-1}(-x)}{\sqrt{2(1+x)}.(-x)}=\frac{\pi}{2\sqrt{2}}$

& for $x\to 0^+ \Rightarrow \{x\}=x$

$\lim\limits_{x\to 0^+} f(x)=\lim\limits_{x\to 0}\frac{\sin^{-1}(1-x)\cos^{-1}(1-x)}{\sqrt{2x}(1-x)}=\frac{\pi}{2}$

42. B, D

Sol. $\because \quad \int_0^{\pi}\sin x dx=2$

$\Rightarrow \quad b-a=4\pi$

$\int_0^{a+b}\cos x dx=9$

$\Rightarrow \quad a+b=\frac{9\pi}{2}$

$\therefore \quad a=\frac{\pi}{4}\&\, b=\frac{7\pi}{4}$

43. B, D

Sol. $\Delta(x)=\begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix}$

$\Rightarrow \Delta'(x)=\begin{vmatrix} f & g & h \\ f' & g' & h' \\ f''' & g''' & h''' \end{vmatrix}$

44. B

Sol. $\because b>\sqrt{ac}$ and $c>\sqrt{bd}$ $(\because$ A.M. > G.M.$)$

$\Rightarrow bc > \sqrt{abcd} \Rightarrow bc > ad$

$\because b > \dfrac{2ac}{a+c}$ and $c > \dfrac{2bd}{b+d}$ $(\because$ A.M. $>$ H.M.$)$

$\Rightarrow bc > \dfrac{4abcd}{(a+c)(b+d)}$

$\Rightarrow (a+c)(b+d) > 4ad$

$\Rightarrow ab+bc+cd > 3ad$

45. A

Sol. $\because \dfrac{a+c}{2} > b$ and $\dfrac{b+d}{2} > c$ $(\because$ A.M. $>$ G.M.$)$

$\Rightarrow a+c > 2b$ and $b+d > 2c$

$\Rightarrow (a+c)+(b+d) > 2(b+c)$

$\Rightarrow a+d > b+c$

$\because b > \dfrac{2ac}{a+c}$ and $c > \dfrac{2bd}{b+d}$ $(\because$ G.M. $>$ H.M.$)$

$\Rightarrow ab+bc+cd > 3ad$

46. C

Sol. $\because \dfrac{ab+cd}{2} > bc$ $(\because$ A.M. $>$ G.M.$)$

$\Rightarrow (ab+cd) > 2bc$

$\because bc > \dfrac{2(ab)(cd)}{(ab+cd)}$ $(\because$ G.M. $>$ H.M.$)$

$\Rightarrow (ab+cd) > 2ad$

47. B
48. A
49. C

SECTION – C

50. 9

Sol. $\dfrac{x-2}{-\frac{1}{\sqrt{2}}} = \dfrac{y-1}{\frac{1}{\sqrt{2}}} = 4\sqrt{2}$ or $-4\sqrt{2}$

$\therefore A(-2, 5)$ & $B(6, -3)$

$\therefore$ area of $\Delta CAB = 20$

$\Rightarrow$ sum $= 2\sqrt{20} = 4\sqrt{5}$

51. 3

Sol. $\because x^2+5x-6=0$

$\Rightarrow (x-1)(x+6)=0 \Rightarrow x=1$

$\therefore x+5y=6$ ..... (i) tangents

& $y = \dfrac{1}{2}(x+1)$ ..... (ii)

$$\therefore \tan\theta = \left|\frac{-\frac{1}{5}+\frac{1}{2}}{1-\frac{1}{10}}\right| = \frac{3}{9} = \frac{1}{3}$$

52. 2

Sol. $\frac{a+b}{2} = \frac{5}{4}\sqrt{ab}$

$\Rightarrow \frac{a}{b} = \frac{1}{4}$

$\frac{H_8 - a}{a - H_8} = 8 \times \frac{a}{b} = 8 \times \frac{1}{4} = 2$

53. 8

Sol. $\because$ $x^2 = -4(y-1)$ has directrix

$y - 1 + 1$

$y = 2$

$\therefore$ $\lambda = 4$

$\mu = -4$

$\lambda - \mu = 8$

54. 2

Sol. $\Delta_1 = 9 - 4a \quad \Rightarrow \quad \Delta_1 + \Delta_2 = a^2 - 4a + 5$

$\Delta_2 = a^2 - 4 \qquad\qquad = (a-2)^2 + 1 > 0$